

Stochastic Constrained Test Assembly for AI-Enabled Assessment Systems

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Abstract

Test assembly, the process of constructing a complete test form from an item pool subject to blueprint constraints, has traditionally been treated as a static optimization problem. In AI-enabled assessment environments, however, item pools evolve continuously as newly generated items enter with uncertain psychometric parameters, and delivery is on demand. These conditions make test assembly a *sequential decision-making problem under uncertainty*: which form should be deployed now, given current but incomplete knowledge of item quality, to simultaneously maximize measurement precision, satisfy content-blueprint constraints, maintain pool sustainability, and accelerate calibration of uncertain new items? This paper proposes the Stochastic Constrained Hybrid (SCH) framework as a principled answer to this question. SCH recasts form-level assembly as a multi-armed bandit (MAB) problem with Fisher information as the reward, extending recent item-level approaches in computerized adaptive testing (CAT) to the form-level setting. A simulation study comparing six test assembly methods is also presented. The main contribution of this paper is a framework for incorporating items with uncertain parameters into the automatic test assembly process for linear test forms.

1 Introduction

Test assembly, the process of constructing a form from an item pool subject to content and psychometric constraints, has traditionally relied on three paradigms: random parallel-form (RPF) assembly (Lord, 1964), Linearly On-the-Fly Testing (LOFT) (Kingsbury and Zara, 1989; Way, 1998, 2025), and constrained optimization via shadow testing (van der Linden, 2005). None was designed for *AI-enabled assessment* environments, where large language models continuously generate items that enter the operational pool with uncertain psychometric parameters and no or little response data.

AI-generated items usually undergo human content validation and many quality controls, but because their empirical psychometric parameters are initially uncertain or imprecise, we need an efficient mechanism for allocating scarce empirical information and constructing tests while those parameters are being learned.

The challenge is direct: items cannot receive reliable difficulty or discrimination estimates until examinees respond, yet they must be deployed immediately to refresh the pool. Traditional solutions—a pilot phase or separate field study—are expensive, create security risks, and artificially separate calibration from administration.

We propose the **Stochastic Constrained Hybrid (SCH) framework**. Drawing on Sharpnack et al. (2024), the core insight is that form assembly can be recast as a *multi-armed bandit* (MAB) problem.

A multi-armed bandit is a sequential decision-making framework in which an agent repeatedly chooses among a set of actions (“arms”), each yielding a stochastic reward, and must balance *exploitation* (choosing the action currently believed to be best) against *exploration* (trying less-certain actions to gather information). The name comes from the image of a row of slot machines (“one-armed bandits”): pulling each lever gives a random payoff, and the agent must learn which lever pays best while minimizing losses during learning. In this paper, each arm is one complete test form—a specific selection of $n = 40$ items satisfying all blueprint constraints—and the reward is the expected measurement precision of that form. Thompson sampling (Thompson, 1933) is a principled algorithm for solving MAB problems: for each bandit arm it maintains a probability distribution over the arm’s true reward, draws a random sample from each distribution, and selects the arm whose sample is highest. Arms whose rewards are highly uncertain receive occasionally large random draws and are therefore explored; arms with well-understood re-

wards are selected only when they genuinely dominate. In the test-assembly context, items with uncertain psychometric parameters (new, insufficiently calibrated items) naturally receive elevated Thompson samples (higher probability to be selected) and are therefore routed to examinees for calibration, without requiring a separate pilot phase.

The SCH framework extends this idea from selecting *one item* per step in a computerized adaptive test (CAT) to selecting a complete *n-item form* subject to a test blueprint specifying content-category quotas and difficulty-band constraints. This is an example of the computational psychometrics approach (von Davier et al., 2021), combining traditional psychometric methods with machine learning algorithms.

The extension of the MAB method from single-item in a computerized adaptive test (CAT, (van der Linden, 1998)) as it was developed by Sharpnack et al. (2024) to *n-item form assembly* is non-trivial for three reasons: (i) the MAB arm space $|\mathcal{C}| = \prod_{\sigma} \binom{|\mathcal{I}_{\sigma}|}{n_{\sigma}} \approx 10^{68}$ is intractable by enumeration, considering the size of the full form-selection arm space being the number of distinct blueprint-feasible test forms that could be assembled from the pool. In this paper we show how to reduce this number to a manageable number of arms per blueprint stratum cell, for a specific, but realistic case, making Thompson sampling computationally feasible. (ii) blueprint constraints couple item choices across content categories and difficulty bands; and (iii) post-selection constraint repair creates a learning inconsistency in the bandit update. SCH resolves all three.

Contributions. (1) A formal extension of bandit item selection to form-level assembly; (2) a stratified cell decomposition (with a pragmatic choice) reducing 10^{68} arms to ≤ 70 per cell while preserving the information-maximization objective; (3) a full delta-method variance approximation over all three IRT parameters, correcting an 88% underestimate from difficulty-only methods; (4) a new information-weighted LOFT baseline revealing a previously undescribed intermediate operating regime; and (5) evidence that pool concentration is a tunable trade-off governed by the exposure-penalty weight.

2 Background and Related Work

IRT and test information. Under the 3PL model, the probability that examinee

j with ability θ_j answers item i correctly is $P_i(\theta) = c_i + (1 - c_i) / (1 + e^{-a_i(\theta - b_i)})$, where $\Phi_i = (a_i, b_i, c_i)$ are discrimination, difficulty, and pseudo-guessing. Fisher information is $I_i(\theta; \Phi_i) = a_i^2 [P_i - c_i]^2 [1 - P_i] / [(1 - c_i)^2 P_i]$. The expected test information of form F is

$$\mathcal{R}(F) = \int \sum_{i \in F} I_i(\theta; \Phi_i) \pi(\theta) d\theta, \quad (1)$$

where $\pi(\theta) \sim \mathcal{N}(0, 1)$ (approximated by 200-point Gaussian quadrature); higher $\mathcal{R}(F)$ means more precise ability estimates.

Blueprint and stratum cells. A blueprint $\mathcal{B}$ specifies form length n , per-category counts $\{n_k\}$ ($\sum_k n_k = n$), difficulty-band bounds, a target mean-difficulty interval $[\underline{b}, \bar{b}]$, and item exposure ceiling $\rho_{\max}$. The $S = K \times L$ intersections of K content categories and L difficulty bands are *stratum cells* σ , each with quota n_{σ} and item pool $\mathcal{I}_{\sigma}$.

Related work. van der Linden (1998) formulated Bayesian CAT item selection. Shadow testing (van der Linden, 2005) added blueprint feasibility via an ILP at each selection step. Sharpnack et al. (2024) reframed CAT selection as a bandit problem (BanditCAT), enabling simultaneous precision and calibration. Sharpnack et al. (2026) added stochastic Symptom–Hetter control (the SAC framework), validated on the Duolingo English Test. This paper extends both from single-item CAT to form-level assembly.

3 Assembly Methods Compared

Stratified Random Parallel Form, **SR** (Lord, 1964) draws n_{σ} items uniformly from each cell item pool $\mathcal{I}_{\sigma}$.

LOFT with exposure control, **LOFT-SH** (Kingsbury and Zara, 1989; Way, 1998, 2025) draws uniformly subject to a Symptom–Hetter (S-H) constraint (Symptom and Hetter, 1985) keeping $\rho_i \leq \rho_{\max}$ via per-item eligibility probabilities α_i updated every W cycles. LOFT that considers the expected information function **LOFT-IW** (new baseline) samples within cells proportional to expected information $\mathcal{R}(\{i\}) = \int I_i(\theta; \Phi_i) \pi(\theta) d\theta$, subject to the same S-H constraint. Shadow test applied to a linear test form **ST-Linear** (van der Linden, 2005) solves one binary ILP per cycle: $\max_{\mathbf{x} \in \{0,1\}^N} \mathbf{v}^{\top} \mathbf{x}$ subject to blueprint constraints, where $v_i = \mathcal{R}(\{i\}) - \lambda_e \rho_i$ ($\lambda_e = 0.5$). **SCH-MAB** and **SCH-EXP** are the new methods that use MAB and they are described in Section 4.

4 The SCH Framework

4.1 Utility Score and Arm Reduction

Each item receives a multi-objective utility score:

$$u_i(t) = \underbrace{\lambda_1 \mathbb{E}_\pi[I_i(\theta; \hat{\Phi}_i)]}_{\text{info (precision)}} + \underbrace{\lambda_2 \text{tr}(\hat{\Sigma}_i)}_{\text{uncertainty (calibration)}} + \underbrace{\lambda_3 \exp(-\alpha(t - t_i^{\text{add}}))}_{\text{refresh (novelty)}} - \underbrace{\lambda_4 \rho_i(t)}_{\text{exposure (sustainability)}}, \quad (2)$$

where $\hat{\Sigma}_i$ is item i 's posterior parameter covariance (calibration uncertainty), t_i^{add} is the administration cycle in which item i entered the pool; $\alpha > 0$ is the novelty decay rate (novelty), and $\rho_i(t)$ its exposure rate (pool health).

Proposition 1 (Reward decomposition). *Under an IRT model, the expected test information of a form decomposes additively over stratum cells: $\mathcal{R}(F) = \sum_\sigma \mathcal{R}(F_\sigma)$ where $F_\sigma = F \cap \mathcal{I}_\sigma$.*

Proof. $\text{TIF}(\theta; F) = \sum_{i \in F} I_i(\theta) = \sum_\sigma \sum_{i \in F_\sigma} I_i(\theta) = \sum_\sigma \text{TIF}(\theta; F_\sigma)$. Integrating against $\pi(\theta)$ and applying linearity of integration gives $\mathcal{R}(F) = \sum_\sigma \mathcal{R}(F_\sigma)$. Hence, linearity of integration over $\pi(\theta)$ proves it. $\square$

In order to make the problem computationally manageable, test form assembly therefore decomposes into S independent per-cell problems. The top- $s=8$ items (pragmatically chosen here) in each cell by u_i form candidate set $\mathcal{Q}_\sigma$; the arm set $\mathcal{A}_\sigma = \binom{\mathcal{Q}_\sigma}{n_\sigma}$ has at most $\binom{8}{4}=70$ arms. Setting $s=8$ satisfies $s \geq 2n_\sigma$ for all $n_\sigma \in \{2, 3, 4\}$ in the $n=40$ blueprint ($K=5, L=3$; item allocation $[10, 8, 8, 8, 6]$).

4.2 Delta-Method Variance for Exploration

For Thompson sampling to explore uncertain items, reward variance must reflect parameter uncertainty:

$$\widehat{\text{Var}}(\mathcal{R}(\mathbf{a})) \approx \sum_{i \in \mathbf{a}} \left[\left(\frac{\partial \mathcal{R}}{\partial a_i} \right)^2 \text{SE}(\hat{a}_i)^2 + \left(\frac{\partial \mathcal{R}}{\partial b_i} \right)^2 \text{SE}(\hat{b}_i)^2 + \left(\frac{\partial \mathcal{R}}{\partial c_i} \right)^2 \text{SE}(\hat{c}_i)^2 \right], \quad (3)$$

where $\mathbf{a}$ is a candidate arm (an n_σ -item subset of $\mathcal{Q}_\sigma$) and $\text{SE}(\hat{a}_i), \text{SE}(\hat{b}_i), \text{SE}(\hat{c}_i)$ are standard errors of the item parameter estimates. A b -only approximation captures only 11.6% of full variance for jump-start items ($\text{SE}(\hat{b}) \approx 0.4$) because $|\partial \mathcal{R} / \partial a_i| \gg |\partial \mathcal{R} / \partial b_i|$: discrimination uncertainty

dominates. Without the full variance, Thompson sampling collapses to pure exploitation and will avoid the new items (exploration).

4.3 SCH-MAB and SCH-EXP

The MAB extension to test form assembly, **SCH-MAB** maintains a Gaussian approximation for each arm $\mathbf{a}$: mean $\mu_{\sigma, \mathbf{a}} = \mathcal{R}(\mathbf{a})|_{\Phi_i = \hat{\Phi}_i}$ and variance $\sigma_{\sigma, \mathbf{a}}^2$ from Eq. (3). It draws $\tilde{\mathcal{R}}_{\sigma, \mathbf{a}} \sim \mathcal{N}(\mu_{\sigma, \mathbf{a}}, \sigma_{\sigma, \mathbf{a}}^2)$ and selects $\mathbf{a}_\sigma^* = \arg \max_{\mathbf{a}} \tilde{\mathcal{R}}_{\sigma, \mathbf{a}}$. Arms with large σ^2 (uncertain items) receive extreme draws and are explored; well-calibrated arms are selected only when their mean dominates. Exposure is controlled through λ_4 in Eq. (2).

MAB with exposure control is the extension of **SCH-EXP** (Sharpnack et al., 2026) from an item level CAT to a linear test that adds Sympon-Hetter (S-H) arm weights: $\mathbf{a}_\sigma^* = \arg \max_{\mathbf{a}} w_{\mathbf{a}} \cdot \max(\tilde{\mathcal{R}}_{\sigma, \mathbf{a}}, 0)^\beta$, where $w_{\mathbf{a}} = \min_{i \in \mathbf{a}} \alpha_i$ and $\beta = 0.5$. This provides a harder exposure ceiling than λ_4 alone.

Feasibility filter. Filtering arms to satisfy the mean-difficulty constraint *before* Thompson sampling eliminates a systematic algorithm learning bias: without it the bandit updates on a repaired arm rather than the one it chose. This improved the constraint satisfaction, CS, from 0.825 to 0.963 for SCH-MAB.

5 Simulation Study

Design. $N=1,000$ items; $n=40$; $K=5$ content categories; $L=3$ difficulty bands; $T=500$ cycles; $J=500$ examinees/cycle; $R=30$ replications; $\rho_{\max}=0.30$. Item parameters: $a_i \sim \text{LogNormal}(0, 0.5)$, $b_i \sim \mathcal{N}(0, 1)$, $c_i \sim \text{Beta}(4, 16)$. New items enter with jump-start calibration ($\text{SE}(\hat{b}_i) \approx 0.30-0.50$). SCH weights: $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (0.40, 0.25, 0.15, 0.20)$ for SCH-MAB; $(0.55, 0.25, 0.15, 0.05)$ for SCH-EXP; $s=8$ for both; Python 3.12, single CPU.

Five scenarios (Table 1) range from static (S1) to inflow plus drift (S5). *Inflow*: new jump-start items added each cycle; *drift*: $b_i + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, 0.05^2)$.

Metrics. Form comparability (FC $\downarrow$): SD of $\mathcal{R}(F_t)$, constraint satisfaction (CS $\uparrow$), average test information (ATI $\uparrow$: mean $\mathcal{R}(F_t)$), pool sustainability via exposure Gini (IPS $\downarrow$), new-item calibration (NIC $\uparrow$: mean $\text{SE}(\hat{b})$ reduction), subgroup fairness via differential test functioning (SF $\downarrow$), and computational cost (CC $\downarrow$, ms/cycle).

Table 1: Simulation scenarios.

ID	Description	Inflow	Drift
S1	Static pool	None	None
S2	Moderate inflow	5%/cycle	None
S3	High inflow	20%/cycle	None
S4	Drift only	None	$\sigma=0.05$
S5	Inflow + drift	10%/cycle	$\sigma=0.05$

6 Results

Table 2 reports cross-scenario means and Figure 1 visualizes the ATI–IPS trade-off. Three distinct operating regimes emerge.

Regime 1 (SR, LOFT-SH). ATI=0.228, IPS=0.177. High pool health, modest precision. SR and LOFT-SH are statistically equivalent on measurement quality; LOFT-SH’s advantage is the provable S-H exposure ceiling. NIC=0.318 arises from new items entering the pool by chance.

Regime 2 (LOFT-IW). ATI=0.350, IPS=0.422. A previously undescribed intermediate regime accessible with a minimal change to any LOFT-SH system: replace uniform within-cell sampling with information-proportional sampling.

Regime 3 (ST-Linear, SCH). ATI $\geq$ 0.64, IPS $\geq$ 0.92. High precision, substantial pool concentration. SCH-MAB matches ST-Linear’s precision (ATI 0.656 vs. 0.642) while providing better constraint satisfaction under drift (CS 0.963 vs. 0.880–0.890) and active new-item calibration (NIC 0.072 vs. 0.039).

Pool concentration. IPS $\approx$ 0.93 means $\approx$ 120 items receive all usage, leaving $\approx$ 880 essentially unused. Sensitivity analysis shows increasing λ_4 from 0.15 to 0.40 reduces IPS by 33% (0.880 $\rightarrow$ 0.586) at an ATI cost of 27% (0.567 $\rightarrow$ 0.411)—concentration is a tunable information–security trade-off, not an algorithmic inevitability. The β temperature in SCH-EXP has negligible effect on IPS ($\beta \in \{0.2, 0.5, 1.0\}$: range 0.812–0.831), confirming that λ_4 , not selection stochasticity, governs concentration.

7 Discussion and Conclusion

This paper introduced the Stochastic Constrained Hybrid (SCH) framework for test form assembly in AI-enabled assessment systems. The motivating question was how to simultaneously maximize measurement precision, satisfy blueprint constraints,

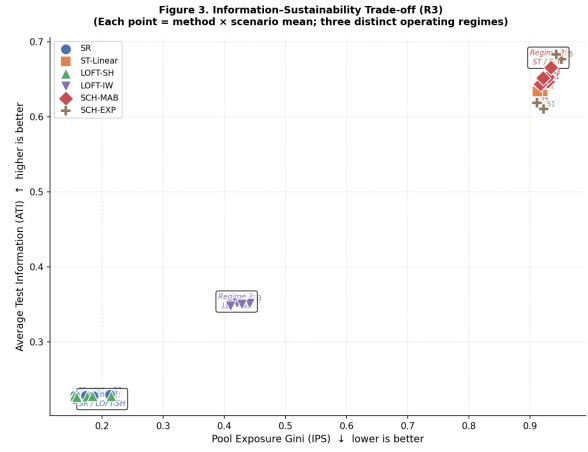


Figure 1: ATI vs. pool exposure Gini (IPS) across six methods and five scenarios. Three operating regimes are clearly separated. LOFT-IW (Regime 2) is a new intermediate regime.

maintain pool sustainability, and accelerate calibration of uncertain new items; four objectives that no existing method addresses jointly. SCH answers this question by recasting form-level assembly as a multi-armed bandit problem, exploiting the additivity of the IRT test information function to decompose an intractable arm problem into small, independent per-cell bandits with a manageable number of arms each.

Three methodological advances distinguish this framework from earlier item-level bandit approaches: a full delta-method variance approximation (correcting an 88% underestimate from difficulty-only approximations); a feasibility filter that ensures the bandit learns from exactly what it assembles; and a theoretical demonstration (confirmed via simulations) that pool concentration is a tunable ATI–security trade-off governed by the exposure-penalty weight λ_4 .

A simulation study comparing six methods reveals three distinct operating regimes. The corrected intermediate LOFT-IW regime (ATI = 0.350, IPS = 0.422) is a new finding with immediate practical implications: it is accessible with minimal modification of any existing LOFT implementation and may serve as the appropriate choice for programs that need more than stratified random sampling but cannot tolerate the pool concentration of full information maximization. The two SCH variants occupy the information-maximizing Regime 3 alongside shadow testing, but offer more stable constraint satisfaction under item-parameter drift and the ability to calibrate uncertain new items

Table 2: Cross-scenario means ($R=30$, $T=500$). Bold = best per metric. [†]NIC: inflow scenarios only. [‡]Cached production cost ≈ 4 ms.

Method	FC↓	CS↑	ATI↑	IPS↓	NIC [†] ↑	SF↓	CC (ms)↓
SR	0.027	0.986	0.228	0.176	0.318	0.007	1.4
ST-Linear	0.016	0.953	0.642	0.921	0.039	0.022	27.9
LOFT-SH	0.027	0.987	0.228	0.178	0.318	0.007	1.7
LOFT-IW [‡]	0.032	0.987	0.350	0.422	0.177	0.012	≈ 4
SCH-MAB	0.014	0.963	0.656	0.930	0.072	0.022	80.6
SCH-EXP	0.062	0.980	0.643	0.926	0.053	0.022	100.5

without a separate pilot phase.

Limitations and future work. Results are simulation-based with very specific default values chosen for pragmatic reasons; real-pool validation is needed. $IPS \approx 0.93$ limits SCH in security-sensitive programs without tuning λ_4 . Natural extensions include MST module routing (Yan et al., 2014, 2024), where modules become bandit arms ($K_g \leq 5$ per stage), and adaptive λ scheduling. More research is needed to evaluate the validity of the test forms assembled with items newly generated items. Similarly, more research is needed to evaluate the tunable exposure rate in real applications.

SCH addresses a genuinely novel problem: assembling blueprint-compliant forms that simultaneously maximize precision, sustain pool health, and calibrate uncertain new items without a pilot phase.

Acknowledgments and Declaration on the Use of Generative AI

This research study was an experiment in using Generative AI as a scientific collaborator, inspired by Google’s Co-Scientist as described in (Gottweis et al., 2025) and other papers such as (Shao et al., 2026). During the preparation of this work, including the exploration of ideas, simulations, drafting, and reference reformatting, I utilized Claude, Sonnet 4.6 (Enterprise). Claude operated with access to Python 3.12 on a single CPU in a Linux (Ubuntu 24) environment. Additionally, DeepSeek-V3 was employed as a reviewer for both the paper and the code. While the primary ideas presented are my own, I interacted extensively with these technologies over several months to iterate, review, and edit all content. I take full responsibility for the final publication.

Finally, I am grateful for the opportunity to have discussed several research papers on random parallel test forms at the 2026 National Council on

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